

AMERICAN AIRCRAFT CARRIER

USS GEORGE WASHINGTON

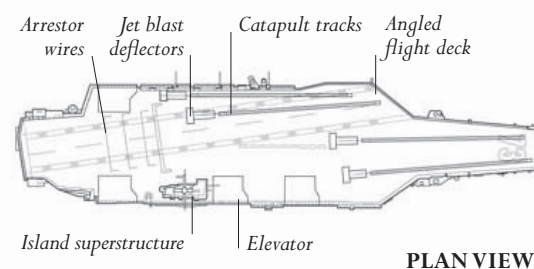
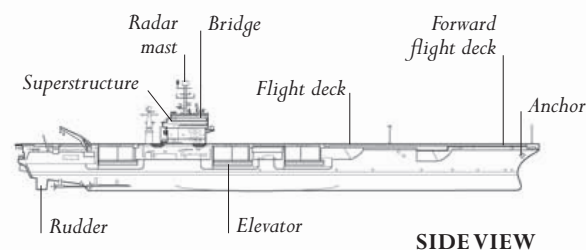
Aircraft carriers are the ultimate symbol of naval power. The *George Washington* is one of 10 Nimitz-class supercarriers in the US Navy – the largest military vessels ever to take to the seas.

Commissioned in 1992, the *George Washington* can accommodate 85 aircraft, including fighter, strike, and transport planes, airborne early warning (AEW) aircraft, and helicopters. It is also a floating home for around 6,000 service men and women. Although designed primarily to offer an offensive strike capability, the ship is equipped with its own defences, such as anti-aircraft and anti-missile weapon systems, and rapid-fire 20mm guns.

On the starboard side and overlooking the deck is the island superstructure – the ship's command-and-control centre, which houses the bridge and primary flight-control area. From here, officers keep a careful watch on the massive flight deck, which covers 1.8 hectares (4.5 acres) – about the size of two-and-a-half soccer pitches.

During flying operations, the deck is a hive of activity, with aircraft taking off and landing, and being manoeuvred, refuelled, and armed. Aircraft are launched by four catapults, two at the forward end of the angled deck and two in the bows. The landing deck is angled to the port side to allow other activities to take place as aircraft return to the ship. When touching down, a pilot must ensure that the plane's tailhook catches one of four high-tensile steel arrestor wires that run across the flight deck. These decelerate the aircraft rapidly and bring it to a halt within two seconds.

When not in use, most aircraft are stored beneath the flight deck in the vast hangar, which stretches for much of the ship's length. On the decks below are living quarters for the ship's personnel.



▲ USS GEORGE WASHINGTON

The sixth Nimitz-class carrier, the *George Washington* is 333m (1,092ft) long and displaces about 103,000 tons. It is powered by two nuclear reactors.

▼ PREPARING FOR TAKE-OFF

Deck crew secure a Northrop Grumman E-2C Hawkeye AEW aircraft to a catapult shuttle.



TAKE-OFF AND LANDING



▲ CATAPULT CONTROL POD

This observation pod, which retracts below the flight deck, enables the crew controlling catapult launches to see what is happening while remaining safe from moving aircraft and their exhausts.



▲ CATAPULT SHUTTLE

For a catapult launch, a shuttle is attached to a plane's undercarriage and propelled along a track by a steam-powered piston.

▲ SHUTTLE TRACK

The catapult shuttle runs along a track set into the surface of the ship's flight deck.



▲ JET-BLAST DEFLECTOR

The yellow deflector shield behind this McDonnell Douglas F/A-18 Hornet prevents the plane's exhaust from causing damage or injury during take-off. The deflector shield is cooled by sea water.



▲ GRUMMAN F-14 TOMCAT LANDING

A pilot increases the throttle at touch-down, so that if the aircraft fails to catch the arrestor wires it still has enough speed to take off again and attempt another landing – a practice called "touch and go".

